

REMARKS ON SIX-TANGENTIAL CONICS TO SMOOTH PLANE SEXTICS

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ABSTRACT.

We work over the complex number field $\mathbb{C}$.

Let B be a smooth plane sextic. A smooth conic Q is called a six-tangential conic to B if Q is tangent to B at distinct 6 points. This paper is motivated by the following simple observation:

Proposition 0.1. *For $\nu = 0, 2, 4, 6$, let $f_\nu(x_0, x_1)$ be a general homogeneous polynomial in variables x_0, x_1 of degree ν . Then the smooth plane sextic B defined by the equation*

$$(0.1) \quad f_6(x_0, x_1) + f_4(x_0, x_1)x_2^2 + f_2(x_0, x_1)x_2^4 + f_0(x_0, x_1)x_2^6 = 0$$

possesses exactly 120 six-tangential conics.

See Section ?? for the proof of this proposition and other beautiful properties of the collection of 120 six-tangential conics.

Let B be a smooth plane sextic. A smooth conic Q is called a splitting conic to B if the local intersection number of B and Q at every point of $B \cap Q$ is even. Let $\pi_B: X_B \rightarrow \mathbb{P}^2$ be the double cover branched along B . Then X_B is a K3 surface. A smooth rational curve on X_B is called a (-2) -curve. Let Q be a splitting conic to B . Then the pull-back of Q under π_B splits into two (-2) -curves $\tilde{Q}^+$ and $\tilde{Q}^-$. Let Q_1 and Q_2 be two distinct splitting conics to B . Then we have

$$\langle \tilde{Q}_1^+, \tilde{Q}_2^+ \rangle \in \{0, 1, 2, 3, 4\}, \quad \langle \tilde{Q}_1^+, \tilde{Q}_2^+ \rangle + \langle \tilde{Q}_1^+, \tilde{Q}_2^- \rangle = 4.$$

Let k be an integer with $k \geq 2$. Let E_k be the set of two-element subsets of $\{1, \dots, k\}$, and let W_k denote the set of mappings

$$w: E_k \rightarrow \{0, 1, 2, 3, 4\}.$$

Thus W_k is the set of weighted complete graphs on k vertices $\{1, \dots, k\}$ with weights $0, 1, 2, 3, 4$. Then the symmetric group $\mathfrak{S}_k$ acts on W_k via its natural action on the vertex set $\{1, \dots, k\}$:

$${}^\tau w(\{i, j\}) := w(\{i^\tau, j^\tau\}).$$

We let the multiplicative group $\{\pm 1\}$ act on $\{0, 1, 2, 3, 4\}$ by

$$a^s := \begin{cases} a & \text{if } s = 1, \\ 4 - a & \text{if } s = -1. \end{cases}$$

Then $\{\pm 1\}^k$ acts on W_k by

$$^{(s_1, \dots, s_k)} w(\{i, j\}) = (w(\{i, j\}))^{s_i s_j}.$$

Thus the semidirect product $G := \{\pm 1\}^k \rtimes \mathfrak{S}_k$ acts on W_k . We call a G -orbit in W_k a splitting graph.

Remark 0.2. The group $\mathfrak{S}_k$ acts on $\{\pm 1\}^k$ by

$$({}^\tau s)_i := s_{i^\tau}.$$

The product in G is given by

$$(s, \tau)(s', \tau') = (s {}^\tau s', \tau \tau').$$

The action of G on W_k is given by

$$^{(s, \tau)} w = s({}^\tau w).$$

Remark 0.3. Our construction is a variant of Shirane's splitting graph adapted to collections of splitting conics.

Suppose that a smooth plane sextic B has a collection Γ of k distinct splitting conics. By labelling $Q_1, \dots, Q_k$ of these splitting conics, and choosing an irreducible component $\tilde{Q}_i^+$ of $\pi_B^{-1}(Q_i)$ for each $i = 1, \dots, k$, we obtain an element w_Γ of W_k by

$$w_\Gamma(\{i, j\}) := \langle \tilde{Q}_i^+, \tilde{Q}_j^+ \rangle.$$

Note that the splitting graph $[w_\Gamma] := G \cdot w_\Gamma$ containing w_Γ is independent of the labelling and of the choices of the irreducible components $\tilde{Q}_i^+$.

Theorem 0.4. *Let γ be a splitting graph in the first column of Table ?? . Then the family $\mathcal{F}_\gamma \subset \mathbb{P}_*(\Gamma(\mathbb{P}^2, \mathcal{O}(6)))$ of smooth plane sextics B possessing a collection Γ of splitting conics such that $[w_\Gamma] = \gamma$ is irreducible, and a general member of $\mathcal{F}_\gamma$ has a collection Γ' of splitting conics such that $[w_{\Gamma'}] = \gamma'$, where γ' is the splitting graph in the same line of Table ?? as γ .*

For example, the first line of Table ?? has the following interpretation. Let B is a general plane sextic in the family of smooth plane sextics possessing two splitting conics Q_1 and Q_2 such that $\langle \tilde{Q}_1^+, \tilde{Q}_2^+ \rangle = 0$ for a suitable choice of the lifts $\tilde{Q}_1^+, \tilde{Q}_2^+$. Then B has another splitting conic Q_3 such that

$$\langle \tilde{Q}_1^+, \tilde{Q}_3^+ \rangle = \langle \tilde{Q}_2^+, \tilde{Q}_3^+ \rangle = 0$$

for a suitable choice of $\tilde{Q}_3^+$.

0.1. Enumeration of splitting conics. Let B be a smooth plane sextic, let $\pi_B: X_B \rightarrow \mathbb{P}^2$ be the double cover branched along B , and let $j: X_B \rightarrow X_B$ be the involution with respect to the double cover $\pi_B: X_B \rightarrow \mathbb{P}^2$. We denote by $h \in \text{Pic}(X_B)$ the class of the pull-back of a line of $\mathbb{P}^2$. Then h is ample. The action on $\text{Pic}(X_B)$ of the involution j leaves h invariant, and acts on its orthogonal complement $h^\perp$ in $\text{Pic}(X_B)$ as the multiplication by -1 . The set $\mathcal{R}_d$ of classes of smooth rational curves C of degree $\langle C, h \rangle = d$ on X_B are enumerated inductively on d as follows:

$$\mathcal{R}_d := \left\{ r \in \text{Pic}(X_B) \mid \begin{array}{l} \langle r, r \rangle = -2, \langle r, h \rangle = d, \\ \forall d' < d \forall r' \in \mathcal{R}_{d'} \langle r, r' \rangle \geq 0 \end{array} \right\}.$$

Then the set of classes of splitting conics for B is equal to

$$\{ r \in \mathcal{R}_2 \mid j^*r \neq r \} / \langle j^* \rangle.$$

0.2. Proof of Proposition 0.1. Let $\iota: \mathbb{P}^2 \rightarrow \mathbb{P}^2$ be the involution given by

$$[x_0 : x_1 : x_2] \mapsto [x_0 : x_1 : -x_2].$$

Let B be a smooth plane sextic defined by an equation of the form (0.1). Then B is ι -invariant. We show that the Picard lattice $\text{Pic}(X_B)$ of X_B of rank 9 and has a basis such that

(a) the Gram matrix of $\text{Pic}(X_B)$ is the diagonal matrix

$$\text{diag}(2, -2, \dots, -2),$$

(b) the class $h \in \text{Pic}(X_B)$ of the pull-back of a line of $\mathbb{P}^2$ is written as

$$h = (3, -1, \dots, -1).$$

Note that the space $\mathcal{I}_\iota$ of ι -invariant sextics is a 15-dimensional subspace of the space $\mathbb{P}_*(\Gamma(\mathbb{P}^2, \mathcal{O}(6)))$ of all plane sextics, and the centralizer of ι in $\text{PGL}(3, \mathbb{C})$ is of dimension 4. Hence the effective moduli of the $K3$ surface X_B is $15 - 4 = 11$. Since B is a general point of $\mathcal{I}_\iota$, the rank of $\text{Pic}(X_B)$ is $20 - 11 = 9$. Let $\phi: \mathbb{P}^2 \rightarrow Q$ be the quotient morphism by the involution ι , where $Q \subset \mathbb{P}^3$ is the quadric surface of rank 3 defined by $z_0 z_2^2 - z_1^2 = 0$, and ϕ is given by

$$[x_0 : x_1 : x_2] \mapsto [z_0 : z_1 : z_2 : z_3] = [x_0^2 : x_1 x_2 : x_1^2 : x_2^2].$$

The fixed point $[0 : 0 : 1]$ of ι is mapped to the apex A of Q and the fixed line $\{x_2 = 0\}$ is mapped to the plane section $Q \cap H_0$ of Q , where $H_0 = \{z_2 = 0\}$. The quotient morphism ϕ induces a double covering $B \rightarrow B_Q$ of a $(2, 3)$ -complete intersection $B_Q \subset Q$. Let $\varpi_B: W_B \rightarrow Q$ be the double covering branched along $B_Q \cup \{A\}$. Then W_B is a del Pezzo surface of degree 1, and the morphism $\varpi_B: W_B \rightarrow Q$ is the bi-anticanonical model of W_B . We denote by α_W the anti-canonical class of W_B .

$\dim_{\mathbb{F}_2} M/I_{1,8}(2)$	number	type of $R(h^\perp)$
0	1	0
1	135	$8A_1$
2	1575	$2D_4$
3	2025	D_8
4	270	E_8

TABLE 0.1. Overlattices M of $I_{1,8}(2)$

The morphism ϕ lifts to a double covering $\varphi: X_B \rightarrow W_B$, and form the following commutative diagram:

$$\begin{array}{ccc} X_B & \xrightarrow{\varphi} & W_B \\ \pi_B \downarrow & & \downarrow \varpi_B \\ \mathbb{P}^2 & \xrightarrow{\phi} & Q. \end{array}$$

The branch locus of the double covering $\varphi: X_B \rightarrow W_B$ is the pull-back $\varpi_B^{-1}(Q \cap H_0)$, which is a member of the bi-anticanonical system $|2\alpha_W|$. Since the pull-back by ϕ of $Q \cap H$ of Q is a conic, we see that $h = \varphi^*\alpha_W$. The del Pezzo surface W_B is obtained by blowing up a projective plane $\mathbf{P}^2$ at distinct 8 points. Hence $\text{Pic}(W_B)$ is isomorphic to $I_{1,8}$, and has a basis $l, e_1, \dots, e_8$, where l is the class of the pull-back of a line of $\mathbf{P}^2$ and $e_1, \dots, e_8$ are the classes of exceptional divisors. We have

$$\alpha_W = 3l - e_1 - \dots - e_8.$$

Now we have

$$I_{1,8}(2) \cong \varphi^*\text{Pic}(W_B) \subset \text{Pic}(X_B).$$

Since $\text{Pic}(X_B)$ is of rank 9, $\text{Pic}(X_B)$ is an overlattice of $I_{1,8}(2)$. By Nikulin's theory of discriminant forms [?], we see that $I_{1,8}(2)$ has $1 + 135 + 1575 + 2025 + 270$ even overlattices as in Table 0.1. Note that

$$h = \varphi^*(3l - e_1 - \dots - e_8)$$

is ample, and hence

$$R(h^\perp) := \{ r \in \text{Pic}(X_B) \mid \langle r, r \rangle = -2, \langle r, h \rangle = 0 \}$$

should be empty. Looking at Table 0.1, we see that the overlattice $\text{Pic}(X_B)$ is in fact equal to $\varphi^*\text{Pic}(W_B)$ and hence has a basis $\varphi^*l, \varphi^*e_1, \dots, \varphi^*e_8$ with the desired properties (a), (b).

Then we see that X_B has exactly 120 splitting conics by the method given in Section 0.1. There are no vectors r in $\text{Pic}(X_B)$ with $\langle r, r \rangle = -2$ and $\langle h, r \rangle = 1$, and there exists exactly 240 vectors r with $\langle r, r \rangle = -2$ and $\langle h, r \rangle = 2$. Therefore these 240 vectors are the classes of smooth rational curves. Moreover the involution j of the double covering $X_B \rightarrow \mathbb{P}^2$ acts on these 240 vectors without fixed points, and

hence the 120 orbits under the involution j are in one-to-one correspondence with splitting conics.

In fact, these 240 vectors r are the pull-back of the 240 lines of the del Pezzo surface W_B . Note that the mapping $B \mapsto B_Q$ gives an isomorphism between the space of ι -invariant sextics and the space $|\mathcal{O}_Q(3)|$. Hence W_B is a general member of the family $\mathcal{U}$ of bi-anticanonical models of del Pezzo surfaces of degree 1 studied in [?]. By [?, Proposition 4.4], the union of 240 lines of W_B has only ordinary double points, which implies that the 120 splitting conics of X_B are indeed six-tangential conics.

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